

WORK INSTRUCTION

Performing Production Data Quality Assessment Using Python

Document ID	WI-DA-001	Version	1.0
Document Type	Work Instruction	Status	Portfolio Simulation
Document Owner	Data Analytics Manager	Intended User	Data Analysts
Related SOP	SOP-DQ-001	Related Specification	SPEC-DQ-001

1. Purpose

This work instruction provides step-by-step instructions for performing the approved initial data quality assessment of manufacturing production datasets using Python. It supports consistent identification, documentation, and disposition of data-quality issues before production data proceed to downstream analysis.

2. Scope

This instruction applies to Data Analysts responsible for initial quality assessment of manufacturing production datasets using the approved Python validation workflow. It covers verification of the working dataset, execution of validation checks, review of results, issue documentation, and determination of dataset disposition.

This instruction does not authorize analysts to modify original source data, delete potential duplicate records, impute missing values, correct source values, or remove statistical anomalies without authorization under the applicable correction or exception process.

3. Prerequisites

- Authorized access to the approved Python/Jupyter analysis environment.
- Authorized access to the production-data project directory.
- An approved working copy of the production dataset.
- Current Production Data Dictionary and applicable validation requirements.
- Access to the validation record and issue-tracking outputs.
- Working knowledge of Python and pandas appropriate to the Data Analyst role.

4. Required Tools and Resources

Tool / Resource	Purpose
Approved Python environment	Executes the controlled validation workflow.
Jupyter Notebook	Runs the approved validation notebook.
pandas	Supports dataset loading and validation operations.
production_data_quality_assessment.ipynb	Approved portfolio-simulation validation notebook.
Production dataset - working copy	Dataset evaluated by the analyst.
Production Data Dictionary	Defines expected fields, types, units, domains, and applicable limits.
Validation requirements	Defines controlled checks and acceptance criteria.
Approved output directory	Stores validation evidence and release outputs.

5. Data Integrity Cautions

CAUTION - SOURCE DATA INTEGRITY

Do not perform validation or cleaning operations on the original source dataset. Perform the assessment using the

authorized working copy. Do not overwrite the source file.

CAUTION - DATA MODIFICATION

Do not automatically delete duplicates, impute missing values, cap range violations, rename source fields, or alter values solely to make a validation check pass.

CAUTION - ANOMALY HANDLING

Do not treat statistical extremity alone as proof that a record is invalid. Preserve the observation and flag it for contextual review.

6. Procedure

Step 1. Verify the Working Dataset

Action: Confirm that the file provided for assessment is the authorized working copy. Verify the dataset identifier, filename, reporting period, and source against the applicable intake record.

Expected result: Dataset identity matches the intake information and the original source dataset remains unchanged.

If not met: Do not proceed. Resolve the discrepancy with the dataset owner or designated responsible person.

Step 2. Open the Approved Validation Notebook

Action: Open Jupyter Notebook in the approved Python environment and navigate to the approved production data quality assessment notebook.

Expected result: The notebook opens without an execution or access error.

If not met: Confirm the approved environment and file location. Escalate unresolved access or environment problems.

Step 3. Verify the Input File Path

Action: Confirm that the notebook input path references the authorized working copy, not the preserved source dataset.

Expected result: The configured path resolves to the intended working-copy file.

If not met: Correct the path only when the authorized file location has been verified. Do not substitute an unverified dataset.

Step 4. Load the Dataset

Action: Execute the dataset-loading section of the notebook.

Expected result: The dataset loads successfully and the notebook reports its row and column dimensions.

If not met: Review the error message. Verify the file path, format, permissions, and approved environment before escalating.

Step 5. Verify Required Structure

Action: Execute the required-field/schema check and compare the dataset structure with the current Production Data Dictionary.

Expected result: All required fields are present.

If not met: Record the applicable schema failure. Follow the severity and disposition requirements; do not fabricate or rename fields without authorization.

Note: The observed number of rows is descriptive unless an approved requirement defines an expected row count.

Step 6. Assess Missing Values

Action: Execute the completeness checks for required identifiers and required measurements.

Expected result: The workflow reports missing counts and applicable validation results for controlled fields.

If not met: Record failures by Rule ID and affected record. Do not independently impute or replace missing values.

Step 7. Identify Potential Duplicate Records

Action: Execute duplicate detection using the approved composite business key: Facility_ID + Batch_ID + Machine_ID + Timestamp.

Expected result: Potential duplicate groups are identified for review.

If not met: Document and investigate the affected records. Do not automatically delete either record.

Step 8. Validate Data Types

Action: Execute data-type validation and compare controlled fields with the expected types defined in the Production Data Dictionary.

Expected result: All evaluated values conform to the required data types.

If not met: Record the field, affected record, observed value, expected type, and Rule ID. Do not guess or silently coerce questionable source values.

Step 9. Validate Controlled Categories

Action: Execute categorical-domain validation for controlled fields such as Facility_ID, Product_Code, Shift_Code, Operator_Team, and Quality_Status.

Expected result: All evaluated categorical values belong to their approved domains.

If not met: Record invalid values and the applicable Rule ID. Do not independently map an invalid source value to an approved category.

Step 10. Validate Numeric Ranges

Action: Execute the configured hard-range validation for controlled numeric measurements.

Expected result: All evaluated numeric values fall within their applicable approved hard validity limits.

If not met: Record each violation. Do not cap, replace, delete, or otherwise alter the source value solely because it fails a range check.

Step 11. Validate Timestamps

Action: Execute timestamp parsing and format validation using the approved timestamp requirement.

Expected result: Required timestamps parse successfully in the approved format.

If not met: Record the affected record and observed value. Do not infer or reconstruct missing/invalid timestamps without authorized source information.

Step 12. Execute Cross-Field Business Rules

Action: Execute applicable relationship checks, including Defect_Count <= Units_Produced and Downtime_Min <= the reporting interval.

Expected result: All applicable cross-field relationships satisfy their controlled requirements.

If not met: Record each failed Rule ID and apply the specified severity. Critical failures place the dataset on HOLD pending investigation.

Step 13. Review Statistical Anomalies

Action: Execute the approved statistical-review check and review observations that exceed the configured review threshold.

Expected result: Potential statistical anomalies are reported separately from hard validation failures.

If not met: Preserve flagged observations and document them for contextual review. Do not automatically delete or fail a record solely because it is statistically unusual.

Step 14. Review the Validation Summary

Action: Review the rule-level validation summary after all applicable checks complete.

Expected result: The summary identifies each executed Rule ID, validation area, severity, failure/flag count, and result.

If not met: If expected rules are absent or execution is incomplete, do not proceed to disposition until the workflow is successfully completed.

Step 15. Review the Detailed Issue Log

Action: Review record-level validation issues and confirm that each issue includes sufficient traceability to the applicable Rule ID and affected record.

Expected result: Issues contain an Issue ID, Rule ID, Record ID, observed condition, expected requirement, severity, and status.

If not met: Do not close or suppress unresolved issues solely to obtain a releasable status.

Step 16. Determine Dataset Disposition

Action: Apply the controlled disposition logic after reviewing the validation summary and detailed issues.

Expected result: The dataset receives HOLD, REVIEW REQUIRED, or APPROVED status according to the applicable criteria.

If not met: Do not override the disposition logic without documented authorization under the applicable exception/change-control process.

Step 17. Preserve Validation Evidence

Action: Save the required validation summary, detailed issue report, release report, and related evidence to the approved output location using the applicable naming convention.

Expected result: All required evidence is present, readable, traceable to the assessed dataset, and retained without overwriting the original source data.

If not met: Do not consider the assessment complete until the required outputs are generated and stored successfully.

7. Dataset Disposition Criteria

Disposition	Condition	Required Response
HOLD	One or more unresolved Critical validation failures are present.	Do not release the dataset for downstream analysis. Investigate and resolve through the applicable process.
REVIEW REQUIRED	No unresolved Critical failures are present, but one or more Major issues require authorized review.	Obtain required review/resolution before release.
APPROVED	Applicable validation is complete and no unresolved Critical or Major issue prevents release.	Record approval and preserve required validation evidence.
REVIEW FLAG	A statistical or contextual condition requires review but does not itself constitute a hard validation failure.	Preserve the observation and complete the required contextual review.

8. Troubleshooting and Exception Handling

Condition	Required Response
Input file not found	Verify the filename and approved working-copy path. Do not substitute an unverified dataset.
Permission denied	Verify the approved location. Contact authorized support if access remains unavailable.
Required Python package unavailable	Confirm the approved environment. Do not independently alter a controlled environment unless authorized.
Notebook execution error	Capture the error. Determine whether it represents a documented validation condition; escalate unresolved execution errors.
Required column missing	Record the schema failure. Do not rename or fabricate fields without authorization.
Potential duplicates detected	Document and investigate. Do not auto-delete.
Required validation output missing	Do not consider the assessment complete. Verify workflow execution and output configuration.
Statistical anomaly identified	Preserve the observation and flag it for contextual review.
Critical validation failure	Record the failure and place the dataset on HOLD according to the applicable requirement.

9. Completion Criteria

The assessment is complete only when all of the following conditions are satisfied:

- All applicable validation checks have executed.
- Required validation outputs have been generated.
- Validation failures and review flags are documented.
- Critical and Major issues have been classified according to applicable requirements.
- Dataset disposition has been determined.
- Required evidence has been saved to the approved location.
- The original source dataset remains unchanged.

10. References

Document ID	Document Title
SOP-DQ-001	Data Quality Validation Procedure
SPEC-DQ-001	Production Data Quality Requirements
FORM-DQ-001	Data Quality Validation Log
Reference	Production Data Dictionary

11. Revision History

Version	Date	Description of Change	Author
1.0	2026-08-19	Initial portfolio release; added visual verification aids and corrected source/working-copy architecture.	Portfolio Author

12. Portfolio Disclaimer

This Work Instruction is a professional portfolio simulation. The company scenario, document identifiers, validation requirements, roles, data, and workflow are fictional and were created to demonstrate technical-

writing, data-quality, instructional-design, and controlled-documentation capabilities. It is not an approved procedure for any real organization.

14. Visual Verification Aids

These figures are supporting aids for the portfolio simulation. The generated validation files remain the authoritative evidence.

Example Validation Summary Output				
Rule_ID	Validation_Area	Severity	Failures_or_Flags	Result
DQ-001	required_fields	Critical	0	PASS
DQ-002	identifier_completeness	Critical	4	FAIL
DQ-004	data_type	Major	4	FAIL
DQ-006	completeness	Major	4	FAIL
DQ-007	potential_duplicates	Major	8	FAIL
DQ-008	numeric_ranges	Major	17	FAIL
DQ-009	allowed_values	Major	8	FAIL
DQ-010	timestamp_validity	Major	4	FAIL
DQ-021	defect_count_consistency	Critical	4	FAIL
DQ-022	downtime_interval_consiste	Critical	4	FAIL
DQ-023	quality_status_consistency	Major	8	FAIL
DQ-024	temperature_anomaly	Review	4	REVIEW

Figure 1. Example rule-level validation summary generated by the workflow.

Example Dataset Release Output	
Release Metric	Result
Records evaluated	24,000
Critical issues	12
Major issues	53
Review flags	4
Release status	HOLD - CRITICAL ISSUE(S) PRESENT

Figure 2. Example dataset release output.

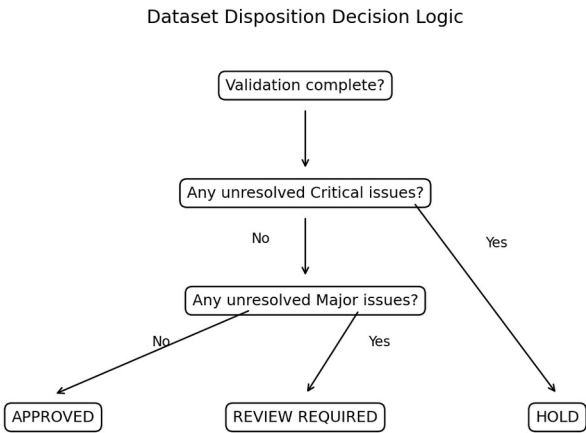


Figure 3. Dataset disposition decision logic.